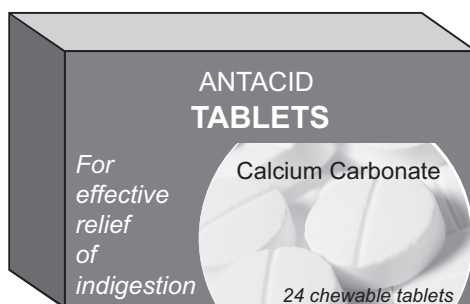


9. Indigestion is caused by excess hydrochloric acid in the stomach. The calcium carbonate in an antacid tablet neutralises the excess acid.

Calcium carbonate reacts with hydrochloric acid according to the equation below.



To determine how much calcium carbonate is present, a crushed indigestion tablet was mixed with water and titrated with dilute hydrochloric acid of concentration 1.0 mol/dm^3 . The end-point was determined using the indicator phenolphthalein. The procedure was repeated three times and the mean volume of hydrochloric acid solution needed to neutralise one tablet was found to be 15.2 cm^3 .

- (a) Calculate the number of moles of hydrochloric acid in 15.2 cm^3 of the 1.0 mol/dm^3 solution. [2]

Number of moles = mol

- (b) Calculate the number of moles of calcium carbonate in one tablet. [1]

Number of moles = mol



- (c) Calculate the mass of calcium carbonate in one tablet. Give your answer in **milligrams**, **mg**. [3]

$$A_r(\text{Ca}) = 40 \qquad A_r(\text{C}) = 12 \qquad A_r(\text{O}) = 16$$

Mass of calcium carbonate = mg

- (d) The diagram shows the labelling on the packet of indigestion tablets.

Each tablet contains:

calcium carbonate	680 mg
magnesium carbonate	80 mg
sucrose and glucose	250 mg
flavouring	
talc	
saccharin	
sodium and magnesium stearate	

Suggest an explanation for the difference between your answer in part (c) and the information given on the packet. [1]

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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



Group

1 H Hydrogen 1																																		
7	Li	9											11	B	12	C	14	N	16	O	19	F	20	Ne										
Lithium	3	Be											Boron	5	Carbon	6	Nitrogen	7	Oxygen	8	Fluorine	9	Neon	10										
23	Na	24											27	Al	28	Si	31	P	32	S	35.5	Cl	40	Ar										
Sodium	11	Mg											Aluminium	13	Silicon	14	Phosphorus	15	Sulfur	16	Chlorine	17	Argon	18										
39	K	40	45	Sc	48	Ti	51	V	52	Cr	55	Mn	56	Fe	59	Co	59	Ni	63.5	Cu	65	Zn	70	Ga	73	Ge	75	As	79	Se	80	Br	84	Kr
Potassium	19	Ca	Scandium	21	Titanium	22	Vanadium	23	Chromium	24	Manganese	25	Iron	26	Cobalt	27	Nickel	28	Copper	29	Zinc	30	Gallium	31	Germanium	32	Arsenic	33	Selenium	34	Bromine	35	Krypton	36
86	Rb	88	89	Y	91	Zr	93	Nb	96	Mo	99	Tc	101	Ru	103	Rh	106	Pd	108	Ag	112	Cd	115	In	119	Sn	122	Sb	128	Te	127	I	131	Xe
Rubidium	37	Sr	Yttrium	39	Zirconium	40	Niobium	41	Molybdenum	42	Technetium	43	Ruthenium	44	Rhodium	45	Palladium	46	Silver	47	Cadmium	48	Indium	49	Tin	50	Antimony	51	Tellurium	52	Iodine	53	Xenon	54
133	Cs	137	139	La	179	Hf	181	Ta	184	W	186	Re	190	Os	192	Ir	195	Pt	197	Au	201	Hg	204	Tl	207	Pb	209	Bi	210	Po	210	At	222	Rn
Caesium	55	Ba	Lanthanum	57	Hafnium	72	Tantalum	73	Tungsten	74	Rhenium	75	Osmium	76	Iridium	77	Platinum	78	Gold	79	Mercury	80	Thallium	81	Lead	82	Bismuth	83	Polonium	84	Astatine	85	Radon	86
223	Fr	226	227	Ac																														
Francium	87	Ra	Actinium	89																														

Key

Key

relative atomic mass

Symbol	Name	Z	A_r
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atomic number